Writing a Simple R Package Using S3 Methods

Building a Table 1 function with S3 dispatch to handle different variable types automatically

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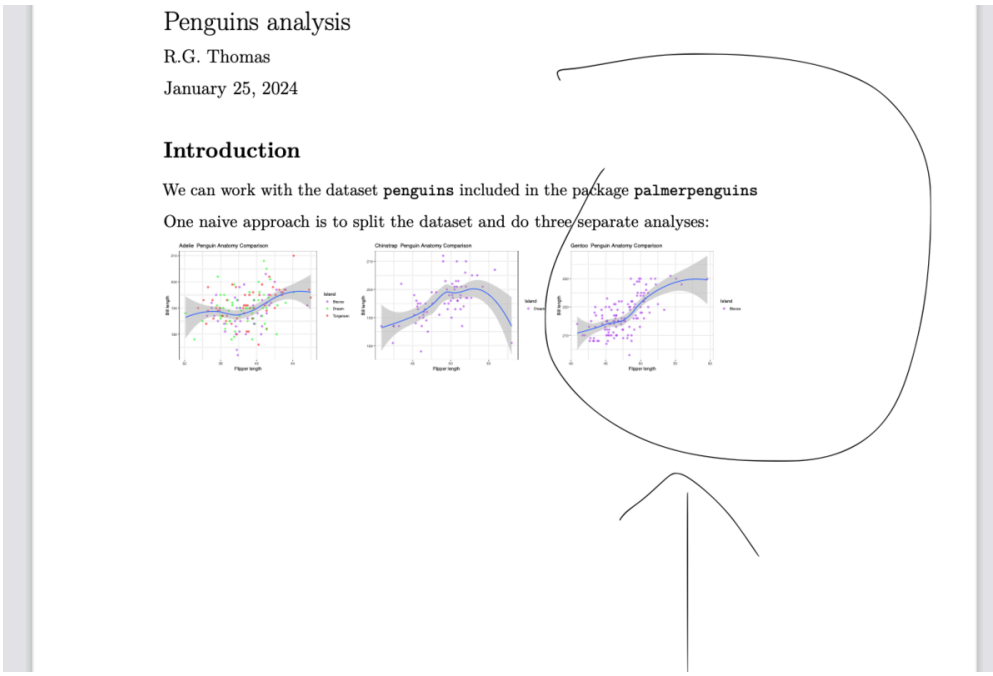

Figure 1: S3 object-oriented programming in R enables functions to behave differently depending on the class of their input.

S3 dispatch lets one function name do the right thing for each variable type.

1 Introduction

[YOUR CONTENT HERE – 2-3 paragraphs. Open with an "I didn't really know…" hook about S3 methods. Describe the problem that led you to learn S3 (e.g., wanting a single function that summarises both numeric and categorical variables for a Table 1). Transition to what this post will accomplish.]

1.1 Motivations

[YOUR CONTENT HERE – 4-6 bullet points. Personal and specific reasons for learning S3. Examples: frustration with if/else chains for variable types, wanting cleaner package code, reading Nick Tierney's paper, etc.]

1.2 Objectives

1. [YOUR CONTENT HERE – specific, measurable]
2. [YOUR CONTENT HERE]
3. [YOUR CONTENT HERE]

4. [YOUR CONTENT HERE]

I am documenting my learning process here. If you spot errors or have better approaches, please let me know.

From scattered functions to a structured package.

2 Prerequisites and Setup

[YOUR CONTENT HERE – 1-2 paragraphs listing what readers need: R 4.0+, devtools, roxygen2, testthat, palmerpenguins. Show library loading code.]

```
library(tidyverse)
library(palmerpenguins)
library(knitr)

set.seed(101)
dat <- palmerpenguins::penguins |>
  filter(!is.na(sex))
```

3 What is S3 Object-Oriented Programming?

[YOUR CONTENT HERE – 1-2 paragraphs in plain language. Define S3 dispatch: when you call a generic function like print(), R looks at the class of the object and calls the appropriate method. Use an analogy. Explain UseMethod(). Transition to the Table 1 use case.]

4 Getting Started: A Simple Generic Function

[YOUR CONTENT HERE – First code example. Show how to create a generic function with UseMethod() and write methods for different classes. Start simple before building the Table 1 function. Follow the CODE -> OUTPUT -> INTERPRETATION pattern.]

```
# [YOUR CODE HERE -- define the generic]
```

[YOUR INTERPRETATION HERE – 2-4 sentences explaining the output.]

```
# [YOUR CODE HERE -- define methods for
#  numeric and factor]
```

[YOUR INTERPRETATION HERE]

Figure 2: A library research environment representing the study and exploration phase of learning a new programming paradigm.

Understanding S3 dispatch is the foundation for writing extensible R packages.

4.1 Building the Table 1 Function

[YOUR CONTENT HERE – The core analysis section. Show how to build a table1() generic that dispatches to methods for numeric variables (mean, SD, range) and categorical variables (counts, percentages). Use the Palmer Penguins dataset as the example.]

```
# [YOUR CODE HERE -- table1 generic + methods]
```

[YOUR INTERPRETATION HERE]

5 Packaging the Functions

[YOUR CONTENT HERE – Show how to organise the S3 methods into an R package structure: R/ directory, DESCRIPTION, NAMESPACE, roxygen2 documentation. Brief walkthrough of devtools::create(), document(), and check().]

```
# [YOUR CODE HERE -- package structure example]
```

[YOUR INTERPRETATION HERE]

6 Checking Our Work

[YOUR CONTENT HERE – Testing the S3 methods. Show testthat tests that verify dispatch works correctly for different variable types. Demonstrate edge cases.]

```
# [YOUR CODE HERE -- testthat examples]
```

[YOUR INTERPRETATION HERE]

6.1 Things to Watch Out For

1. **[YOUR GOTCHA HERE]** – [explanation]
2. **[YOUR GOTCHA HERE]** – [explanation]
3. **[YOUR GOTCHA HERE]** – [explanation]
4. **[YOUR GOTCHA HERE]** – [explanation]
5. **[YOUR GOTCHA HERE]** – [explanation]

Figure 3: A tidy desk at the end of a work session, representing the completion of a structured programming project.

A well-structured S3 system makes future extensions straightforward.

6.2 Lessons Learnt

6.2.1 Conceptual Understanding

- [YOUR CONTENT HERE – what you learnt about S3 dispatch and OOP in R]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.2.2 Technical Skills

- [YOUR CONTENT HERE – tools/packages mastered: UseMethod(), roxygen2, devtools, etc.]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.2.3 Gotchas and Pitfalls

- [YOUR CONTENT HERE – practical warnings for future S3 developers]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.3 Limitations

- [YOUR CONTENT HERE – 5-6 bullets. Cover: S3 has no formal class definitions, no input validation, limited compared to S4/R6, Table 1 function scope, dataset limitations, etc.]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.4 Opportunities for Improvement

1. [YOUR CONTENT HERE – extensions to try: S4/R6 comparison, CRAN submission, additional methods, etc.]
2. [YOUR CONTENT HERE]
3. [YOUR CONTENT HERE]
4. [YOUR CONTENT HERE]
5. [YOUR CONTENT HERE]
6. [YOUR CONTENT HERE]

7 Wrapping Up

[YOUR CONTENT HERE – 2-3 paragraphs. First: recap the main accomplishment. Second: what you learnt personally. Third: advice for readers.]

Main takeaways:

- [YOUR CONTENT HERE – 3-4 quantitative or specific results]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

8 See Also

Related posts:

- [YOUR RELATED POST 1]
- [YOUR RELATED POST 2]

Key resources:

- A Simple Guide to S3 Methods (Tierney) – The ArXiv paper that inspired this post
- Advanced R: S3 (Wickham) – Comprehensive S3 reference
- R Packages (Wickham & Bryan) – The definitive guide to R package development
- devtools documentation – Tools for R package development
- Palmer Penguins Dataset – The dataset used in the examples

9 Reproducibility

[YOUR CONTENT HERE – environment description, R version, key package versions.]

File	Purpose
[YOUR FILES HERE]	[descriptions]

```
# To reproduce:
# [YOUR PIPELINE COMMANDS HERE]
```

10 Let's Connect

- **GitHub:** rgt47
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- **LinkedIn:** Ronald Glenn Thomas
- **Email:** rgtlab.org/contact

I would enjoy hearing from you if:

- You spot an error or a better approach to any of the code in this post.
- You have suggestions for topics you would like to see covered.
- You want to discuss R programming, data science, or reproducible research.
- You have questions about anything in this tutorial.
- You just want to say hello and connect.